IT & CYBER
INTERNSHIP PROGRAM

6-Week

Foundation + Specialization Learning Path

20 Interns · 5 Specialization Tracks · 2-Week Common Foundation
NOC · SOC · Cloud (OpenStack) · Red Team / Pentesting · Cyber R&D

Curriculum · Hands-on labs · Weekly KPIs · Capstone Projects

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1. Program Overview

This roadmap lays out a **6-week, two-phase learning path** for **interns** entering a IT & Cyber Internship Program. It is built in the visual style of colour-coded topic nodes, clear prerequisite flow, hands-on labs at every step, and curated external resources for deeper learning.

Field	Detail
Total Duration	6 weeks (2 weeks Common Foundation + 4 weeks Specialization)
Total Interns	20
Target Audience	Final-year university students or fresh graduates — beginner to intermediate technical level
Program Goal	Prepare interns for real-world industry roles through structured learning, extensive hands-on labs, and project-based learning
Learning Model	Phase 1: shared curriculum for all interns → Phase 2: 5 specialization tracks selected by fit & interest
Assessment Model	Daily labs, weekly quizzes, weekly KPI scorecards, mini-projects, and a final capstone per track (see Section 9)

1.1 Program Structure at a Glance

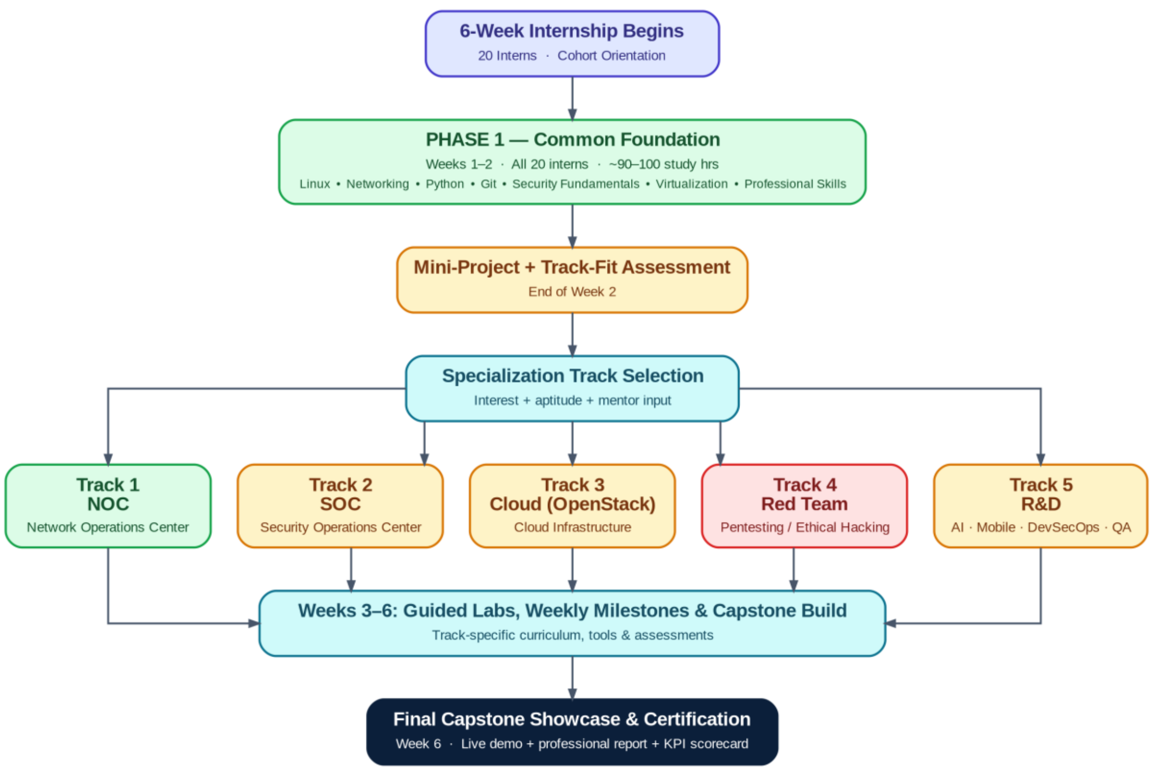

Figure 1 — End-to-end program flow: Common Foundation → Track Selection → Specialization → Capstone Showcase

1.2 The Two Phases

Phase 1 — Common Foundation (Weeks 1–2)

All 20 interns follow an identical curriculum covering Linux, networking, Python, Git, cybersecurity fundamentals, virtualization, and professional skills. This ensures every intern — regardless of specialization — shares a common technical baseline before splitting into tracks. Phase 1 ends with a graded mini-project and a track-fit conversation with mentors.

Phase 2 — Specialization (Weeks 3–6)

Interns split into five tracks aligned to real industry job families:

- **Track 1 — NOC** (Network Operations Center)
- **Track 2 — SOC** (Security Operations Center)
- **Track 3 — Cloud** (OpenStack-based private cloud engineering)
- **Track 4 — Cybersecurity** (Red Team / Penetration Testing)
- **Track 5 — R&D** (Research & Development — AI, Mobile, DevSecOps, or Automation)

Each track runs for 4 weeks with its own learning objectives, weekly milestones, hands-on labs, mini-projects, required tools, and a final capstone project mapped to specific job roles.

1.3 How to Read This Roadmap

Every topic node throughout this document follows the same format, so you can scan quickly: a coloured difficulty marker, a mandatory/optional badge, an estimated study-time budget, and (where relevant) a prerequisite. A worked example:

● INTERMEDIATE MANDATORY ~14h Prereq: Linux Fundamentals

Marker	Meaning
● Beginner	No prior experience assumed beyond general computer literacy.
● Intermediate	Builds directly on Beginner topics earlier in the same track or Phase 1.
● Advanced	Requires comfort with the Intermediate material that precedes it; expect a steeper learning curve.
MANDATORY	Required for every intern in that track to reach capstone readiness.
OPTIONAL	Stretch content for interns who finish core material early or want deeper specialization.
~Nh	Estimated self-study + lab hours to reach working competency (excludes live instruction).
Prereq:	Topic(s) that should be completed first; "None" means it can start immediately.

Note on resources: to keep this roadmap usable rather than overwhelming, resources are grouped at the topic-cluster level rather than one link per micro-topic. Every cluster lists official documentation first, then free courses, video walkthroughs, hands-on practice platforms, repositories, and — where useful — books or certifications.

2. Phase 1 — Common Foundation (Weeks 1–2)

All 20 interns complete this identical two-week curriculum before selecting a specialization track. Topics build progressively — operating systems and networking first, then programming and version control, then security fundamentals and virtualization — so that every concept has the prerequisite knowledge it needs already in place.

2.1 Learning Objectives

- Operate confidently in a Linux command-line environment, including permissions, processes, and basic Bash automation.
- Explain and apply core networking concepts (OSI/TCP-IP, DNS/DHCP, routing/switching, VPNs) and use Wireshark to inspect real traffic.
- Write Python scripts that automate real tasks and interact with REST APIs.
- Use Git/GitHub professionally: branches, pull requests, and code review collaboration.
- Explain the CIA triad, authentication vs. authorization, encryption, hashing, and PKI — and apply basic vulnerability-management practice.
- Provision and use virtual machines and Docker containers as disposable lab environments.
- Produce clear technical documentation and work inside a lightweight Agile/Scrum cadence.

2.2 Foundation Roadmap

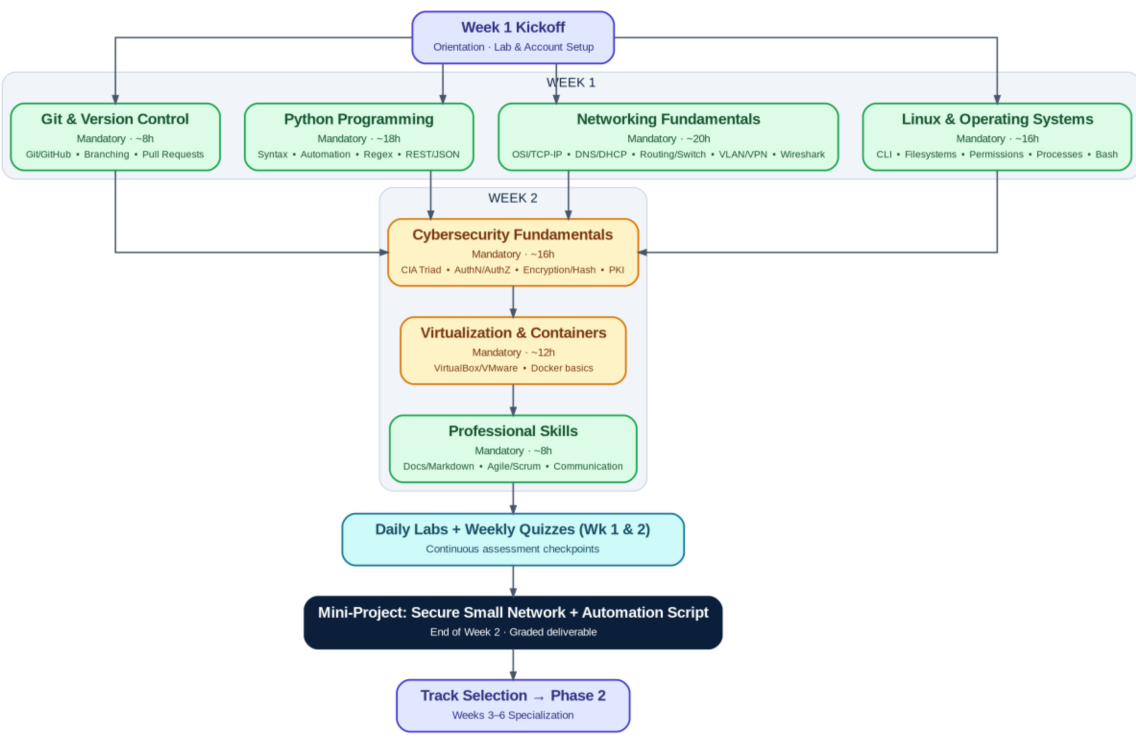

Figure 2 — Phase 1 topic flow across Week 1 and Week 2, converging on the Week 2 mini-project and track selection

2.3 Topic Breakdown

2.3.1 Operating Systems — Linux

● **BEGINNER** **MANDATORY** **~16h** *Prereq: None*

The command line is the foundation every other Phase 1 topic (and every track) is built on.

Covers: Linux fundamentals • Linux command line • File systems • User & permission management • Process management • Bash scripting

Hands-on labs:

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Resources:

- **Official Docs:** Ubuntu Documentation
- **Free Course:** Linux Foundation — Introduction to Linux (LFS101)
- **Video / YouTube:** freeCodeCamp — Linux Crash Course (with labs)
- **Hands-on / Practice:** OverTheWire: Bandit wargame
- **Book (optional):** The Linux Command Line (W. Shotts, free PDF)
- **Certification (optional):** CompTIA Linux+

2.3.2 Networking Fundamentals

● **BEGINNER** **MANDATORY** **~20h** *Prereq: Linux (helps, not required)*

Covers the full stack from cabling to application-layer protocols, with Wireshark tying theory to real packets.

Covers: OSI Model • TCP/IP • IPv4 & IPv6 • DNS • DHCP • Routing • Switching • VLANs • VPNs • HTTP/HTTPS • SSH • SMTP • FTP • Firewalls • Wireshark

Hands-on labs:

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Resources:

- **Official Docs:** Cisco Networking Academy — Skills for All
- **Free Course:** freeCodeCamp — How Computer Networks Work (9-hr, Network+ aligned)
- **Video / YouTube:** freeCodeCamp.org YouTube channel
- **Hands-on / Practice:** Wireshark Sample Captures library
- **Certification (optional):** CompTIA Network+

2.3.3 Programming — Python

● **BEGINNER** **MANDATORY** **~18h** *Prereq: None*

Python is the automation language used across every track later in the program — this cluster gets everyone to a working baseline.

Covers: Python fundamentals • Python for automation • Regular expressions • APIs • JSON • REST

Hands-on labs:

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Resources:

- **Official Docs:** Python Official Documentation
- **Free Course:** freeCodeCamp — Python Full Course for Beginners
- **Video / YouTube:** freeCodeCamp — Networking in Python (4 hands-on projects)
- **Hands-on / Practice:** Real Python tutorials
- **Certification (optional):** PCEP — Certified Entry-Level Python Programmer (Python Institute)

2.3.4 Version Control — Git & GitHub

● **BEGINNER** **MANDATORY** **~8h** *Prereq: None*

Every deliverable from here forward — labs, mini-projects, capstones — is submitted through Git, so this cluster is scheduled early.

Covers: Git • GitHub • Branching • Pull Requests • Collaboration

Hands-on labs:

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Resources:

- **Official Docs:** Git Official Documentation
- **Free Course:** GitHub Docs — Get Started
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Git and GitHub for Beginners"
- **Hands-on / Practice:** GitHub Skills (interactive courses)
- **Repository:** first-contributions (beginner PR practice repo)

2.3.5 Cybersecurity Fundamentals

● **INTERMEDIATE** **MANDATORY** **~16h** *Prereq: Networking Fundamentals*

The shared security vocabulary and mental model every track — NOC, SOC, Cloud, Red Team, R&D — builds on.

Covers: CIA Triad • Authentication • Authorization • Encryption • Hashing • PKI • Vulnerability Management • Security Best Practices

Hands-on labs:

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Resources:

- **Official Docs:** NIST Cybersecurity Framework (CSF 2.0)
- **Free Course:** Cisco — Introduction to Cybersecurity (Skills for All)
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Cyber Security Full Course"
- **Hands-on / Practice:** TryHackMe — "Pre Security" learning path
- **Certification (optional):** CompTIA Security+

2.3.6 Virtualization & Containers

● **INTERMEDIATE** **MANDATORY** **~12h** *Prereq: Linux*

Every lab from Week 2 onward runs inside a VM or container, so interns need to be comfortable standing environments up and tearing them down.

Covers: VirtualBox • VMware • Docker fundamentals

Hands-on labs:

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Resources:

- **Official Docs:** Docker Official Documentation
- **Free Course:** Docker — Get Started Guide
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Docker Tutorial for Beginners"
- **Hands-on / Practice:** Play with Docker (browser-based sandbox)

2.3.7 Professional Skills

● **BEGINNER** **MANDATORY** **~8h** *Prereq: None*

Technical skill without communication skill doesn't ship — this cluster is graded as part of every later deliverable, not just Phase 1.

Covers: Documentation • Technical writing • Markdown • Agile • Scrum • Communication • Presentation skills

Hands-on labs:

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Resources:

- **Official Docs:** Markdown Guide
- **Free Course:** Atlassian Agile Coach — Agile & Scrum 101
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Technical Writing"

2.4 Week 1 Daily Schedule

Day	Morning Focus	Afternoon Focus	Daily Lab / Deliverable
Mon	Orientation, account & lab-environment setup, cohort introductions	Linux basics: shell navigation, filesystem hierarchy	Document a Linux filesystem walkthrough (CLI-only)
Tue	Linux permissions & ownership	Process management + intro Bash scripting	Bash script: bulk-fix a broken permission tree
Wed	Networking: OSI model, TCP/IP, IPv4/IPv6	DNS, DHCP, routing & switching basics	Wireshark capture: identify 6+ protocols
Thu	VLANs, VPNs, firewalls, HTTP/HTTPS, SSH	Python fundamentals begin	Packet Tracer: build a mini routed/switched topology
Fri	Python for automation & regular expressions	Git & GitHub setup, first branch + pull request	Weekly Quiz 1 + automation script submitted via PR

2.5 Week 2 Daily Schedule

Day	Morning Focus	Afternoon Focus	Daily Lab / Deliverable
Mon	APIs, REST & JSON in Python	Cybersecurity fundamentals: CIA triad, authN/authZ	Script: consume a public REST API end-to-end
Tue	Encryption, hashing, PKI	Vulnerability management & security best practices	Hash-verification exercise + guided vulnerability scan
Wed	Virtualization: VirtualBox / VMware	Docker fundamentals	Provision 2 VMs + containerize a script
Thu	Professional skills: documentation, Markdown, Agile/Scrum	Mini-project build session (mentor office hours)	README + sprint board set up for mini-project
Fri	Mini-project completion & rehearsal	Mini-project demo day + track-fit conversations	Weekly Quiz 2 + Mini-Project live demo

2.6 Weekly Assessment Checkpoints

Checkpoint	Covers	Format	Weight toward Foundation KPI
Daily Labs (10 total)	That day's topic cluster	Graded pass/revise checklist, submitted via Git	40%
Weekly Quiz 1 (Fri, Wk 1)	Linux, Networking, Python basics, Git	20 MCQ/short-answer, 30 minutes	15%
Weekly Quiz 2 (Fri, Wk 2)	Security fundamentals, virtualization, professional skills	20 MCQ/short-answer, 30 minutes	15%
Mini-Project (end of Wk 2)	Integrates all 7 clusters	Working deliverable + live demo + README	30%

Full scoring rubrics and the printable Weekly KPI Scorecard used to record these checkpoints are provided in Section 9 — Intern KPI & Evaluation Templates.

2.7 Mini-Project — End of Week 2

"Secure Small Network + Automation Toolkit"

Working in pairs, interns design and document a small routed/switched lab network (Packet Tracer or GNS3), harden one Linux host on it (SSH hardening, firewall rules, least-privilege users), and write a Python automation script that checks the host's security posture (e.g., verifies firewall rules, flags weak permissions, or polls a REST API health check). The deliverable is submitted as a GitHub repository with a full README, and presented as a 5-minute live demo to mentors.

Required deliverables: (1) network topology diagram, (2) hardened VM with documented configuration, (3) working Python automation script with tests, (4) README.md, (5) 5-minute live demo.

Evaluation: scored against the Mini-Project Evaluation Rubric in Section 9.4, and feeds directly into each intern's track-fit conversation with mentors.

10. Appendices

Appendix A — Consolidated Master Resource List

Every resource cited throughout this roadmap, grouped by category for quick reference. Track-specific context for each link is in its topic cluster in Sections 2 and 4–8.

Official Documentation

Resource	URL
Python	https://docs.python.org/3/
Git	https://git-scm.com/doc
GitHub Docs	https://docs.github.com/en/get-started
Docker	https://docs.docker.com/
Ubuntu	https://help.ubuntu.com/
Cisco Networking Academy	https://www.netacad.com/
Wireshark	https://www.wireshark.org/docs/
NIST Cybersecurity Framework	https://www.nist.gov/cyberframework
NIST CSRC	https://csrc.nist.gov/
MITRE ATT&CK	https://attack.mitre.org/matrices/enterprise/
OWASP Top 10	https://owasp.org/Top10/
Wazuh	https://documentation.wazuh.com/current/index.html
Elastic (ELK Stack)	https://www.elastic.co/guide/index.html
Splunk	https://docs.splunk.com/Documentation/Splunk
SigmaHQ	https://github.com/SigmaHQ/sigma
OpenStack	https://docs.openstack.org/
Kolla Ansible	https://docs.openstack.org/kolla-ansible/latest/
DevStack	https://docs.openstack.org/devstack/latest/
Zabbix	https://www.zabbix.com/documentation/
Grafana	https://grafana.com/docs/
LibreNMS	https://docs.librenms.org/
Prometheus	https://prometheus.io/docs/
HashiCorp Terraform	https://developer.hashicorp.com/terraform
Ansible	https://docs.ansible.com/
Nmap Reference Guide	https://nmap.org/book/man.html
BloodHound	https://bloodhound.specterops.io/

Resource	URL
Metasploit	https://docs.metasploit.com/
GTFOBins	https://gtfobins.github.io/
John the Ripper	https://www.openwall.com/john/
NetExec (formerly CrackMapExec)	https://github.com/Pennyw0rth/NetExec
Android Developers	https://developer.android.com/
Kotlin	https://kotlinlang.org/docs/home.html
Firebase	https://firebase.google.com/docs
Apple Developer Documentation	https://developer.apple.com/documentation/
OpenAI API	https://platform.openai.com/docs
LangChain	https://python.langchain.com/
Hugging Face	https://huggingface.co/
Kubernetes	https://kubernetes.io/docs/home/
Trivy	https://github.com/aquasecurity/trivy
Selenium	https://www.selenium.dev/documentation/
Playwright	https://playwright.dev/
PowerShell (Microsoft Learn)	https://learn.microsoft.com/en-us/powershell/
Sysmon (Sysinternals)	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
Markdown Guide	https://www.markdownguide.org/

Free Courses & Learning Platforms

Resource	URL
Linux Foundation — Introduction to Linux (LFS101)	https://training.linuxfoundation.org/training/introduction-to-linux/
freeCodeCamp	https://www.freecodecamp.org/
Cisco Skills for All	https://www.netacad.com/
PortSwigger Web Security Academy	https://portswigger.net/web-security
roadmap.sh	https://roadmap.sh/
roadmap.sh — DevSecOps	https://roadmap.sh/devsecops
roadmap.sh — AI Engineer	https://roadmap.sh/ai-engineer
roadmap.sh — Android	https://roadmap.sh/android
Atlassian Agile Coach	https://www.atlassian.com/agile
GitHub Skills	https://skills.github.com/
Real Python	https://realpython.com/

Hands-On Practice Platforms

Resource	URL
OverTheWire — Bandit wargame	https://overthewire.org/wargames/bandit/
TryHackMe	https://tryhackme.com/
HackTheBox Academy	https://academy.hackthebox.com/catalogue/paths
LetsDefend	https://letsdefend.io/
MITRE ATT&CK Navigator	https://mitre-attack.github.io/attack-navigator/
Play with Docker	https://labs.play-with-docker.com/
OSINT Framework	https://osintframework.com/
Wireshark Sample Captures	https://wiki.wireshark.org/SampleCaptures
VirusTotal	https://www.virustotal.com/

Repositories

Resource	URL
first-contributions (beginner PR practice)	https://github.com/firstcontributions/first-contributions
openstack/kolla-ansible	https://github.com/openstack/kolla-ansible
BloodHound	https://github.com/SpecterOps/BloodHound
THC-Hydra	https://github.com/vanhauser-thc/thc-hydra

Resource	URL
NetExec	https://github.com/Pennyw0rth/NetExec
Trivy	https://github.com/aquasecurity/trivy

Certifications (Optional)

Certification	URL
CompTIA (Linux+, Network+, Security+)	https://www.comptia.org/
Cisco CCNA	https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna/index.html

Certifications are optional stretch goals, not program requirements. Interns interested in pursuing one should discuss timing with their mentor — most are best attempted after, not during, the internship.

Appendix B — Required Tools & Software Checklist

For lab-environment setup ahead of Day 1. All tools are free / open-source or have a free tier sufficient for this program.

Phase 1 — All Interns

Tool	Used For
VirtualBox or VMware Workstation Player	Local VM hosting for Linux/Docker labs
A Linux distribution ISO (e.g., Ubuntu Server/Desktop)	Primary Linux lab environment
Wireshark	Packet capture and analysis
Cisco Packet Tracer	Basic networking topology labs
Python 3.x + pip	Programming and automation labs
Git + a GitHub account	Version control for every deliverable
Docker Desktop / Docker Engine	Containerization labs
A code editor (e.g., VS Code)	Writing scripts, configs, and documentation

Track 1 — NOC

- Cisco Packet Tracer
- GNS3 (+ a router/switch image set)
- Wireshark
- Zabbix
- Grafana
- LibreNMS

Track 2 — SOC

- Wazuh
- ELK Stack (Elasticsearch, Logstash, Kibana)
- Splunk (free tier)
- Microsoft Sentinel (trial/lab tenant)
- Velociraptor
- Sysmon
- VirusTotal account

Track 3 — Cloud (OpenStack)

- DevStack-compatible Linux VM (min. 8GB RAM recommended)
- Kolla-Ansible
- OpenStack CLI (python-openstackclient)
- Terraform
- Ansible
- Prometheus + Grafana

Track 4 — Red Team

- Kali Linux (VM or bare metal)
- Burp Suite Community Edition
- Metasploit Framework
- BloodHound + Neo4j
- NetExec
- Gobuster
- Hydra
- John the Ripper

Track 5 — R&D

- Domain-dependent — see below

R&D domain-specific tools: AI/LLMs — Python, LangChain, Hugging Face account, OpenAI-compatible API key. Android — Android Studio, Kotlin. iOS — Xcode (macOS required), Swift. DevSecOps — GitHub Actions or Jenkins, Docker, Kubernetes (e.g., kind/minikube), Trivy, SonarQube (Community Edition). Automation — Selenium or Playwright, PowerShell.

Appendix C — Glossary of Acronyms

Term	Meaning
ACL	Access Control List — rules that permit or deny traffic
AD	Active Directory — Microsoft's directory service for Windows networks
API	Application Programming Interface
ATT&CK	Adversarial Tactics, Techniques & Common Knowledge (MITRE framework)
CIA Triad	Confidentiality, Integrity, Availability — core security model
CI/CD	Continuous Integration / Continuous Deployment
CLI	Command-Line Interface
CVSS	Common Vulnerability Scoring System
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
EDR	Endpoint Detection & Response
ELK	Elasticsearch, Logstash, Kibana (log analytics stack)
HA	High Availability
HOT	Heat Orchestration Template (OpenStack)
IaC	Infrastructure as Code
IaaS	Infrastructure as a Service
IOC	Indicator of Compromise
IR	Incident Response
IPsec	Internet Protocol Security (VPN encryption standard)
LLM	Large Language Model
MTTD / MTTR	Mean Time to Detect / Mean Time to Respond (or Repair)
MVVM	Model-View-ViewModel (software architecture pattern)
NAT / PAT	Network Address Translation / Port Address Translation
NOC	Network Operations Center
NSE	Nmap Scripting Engine
OSI Model	Open Systems Interconnection model (7-layer networking reference)
OSINT	Open-Source Intelligence
OSPF	Open Shortest Path First (routing protocol)
PKI	Public Key Infrastructure
PTES	Penetration Testing Execution Standard
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer (API architecture style)

Term	Meaning
SIEM	Security Information & Event Management
SNMP	Simple Network Management Protocol
SOAR	Security Orchestration, Automation & Response
SOC	Security Operations Center
SSH	Secure Shell
STP	Spanning Tree Protocol
TCP/IP	Transmission Control Protocol / Internet Protocol
VLAN	Virtual Local Area Network
VPN	Virtual Private Network